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TPL A2

Praktikum 1

```python
matrix = [
# 0  1  2  3
 [0, 1, 1, 0],  # Baris 0
 [1, 0, 0, 1],  # Baris 1
 [1, 0, 0, 1],  # Baris 2
 [0, 1, 1, 0]   # Baris 3
]

# Tampilkan matrix
print("Adjacency Matrix:")
for row in matrix:
    print(row)

print("\nPenjelasan setiap baris:")

# Penjelasan tiap baris
# Baris 0
print("Baris 0 [0,1,1,0] -> Node 0 terhubung ke node 1 dan 2")

# Baris 1
print("Baris 1 [1,0,0,1] -> Node 1 terhubung ke node 0 dan 3")

# Baris 2
print("Baris 2 [1,0,0,1] -> Node 2 terhubung ke node 0 dan 3")

# Baris 3
print("Baris 3 [0,1,1,0] -> Node 3 terhubung ke node 1 dan 2")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
> & C:/Users/ASUS/AppData/Local/Programs/Python/Python313/pyth
Adjacency Matrix:
[0, 1, 1, 0]
[1, 0, 0, 1]
[1, 0, 0, 1]
[0, 1, 1, 0]

Penjelasan setiap baris:
Baris 0 [0,1,1,0] -> Node 0 terhubung ke node 1 dan 2
Baris 1 [1,0,0,1] -> Node 1 terhubung ke node 0 dan 3
Baris 2 [1,0,0,1] -> Node 2 terhubung ke node 0 dan 3
Baris 3 [0,1,1,0] -> Node 3 terhubung ke node 1 dan 2
PS D:\sem 2\algoritma\Praktikum 11>
```

Praktikum 2

```python
# Graph:
# A --- B
# |     |
# C --- D

# Dictionary digunakan untuk menyimpan adjacency list
# Key = node
# Value = list tetangga (node yang terhubung)
graph = {
    "A": ["B", "C"],  # A terhubung ke B dan C
    "B": ["A", "D"],  # B terhubung ke A dan D
    "C": ["A", "D"],  # C terhubung ke A dan D
    "D": ["B", "C"]   # D terhubung ke B dan C
}

# Menampilkan adjacency list
print("Adjacency List:")
for node, tetangga in graph.items():
    # node = simpul saat ini
    # tetangga = daftar simpul yang terhubung
    print(node, "->", tetangga)

# Penjelasan:
# A -> ['B', 'C'] artinya A terhubung ke B dan C
# B -> ['A', 'D'] artinya B terhubung ke A dan D
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\sem 2\algoritma\Praktikum 11> & C:/Users/ASUS/AppData/Local/Programs/
Adjacency List:
A -> ['B', 'C']
B -> ['A', 'D']
C -> ['A', 'D']
D -> ['B', 'C']
PS D:\sem 2\algoritma\Praktikum 11>
```

Praktikum 3

```python
matrix = [
    [0,1,1,0],  # A
    [1,0,1,0],  # B
    [1,1,0,1],  # C
    [0,0,1,0]   # D
]

# Nama node (sesuai urutan matrix)
nodes = ["A", "B", "C", "D"]

# Dictionary kosong untuk menyimpan hasil adjacency list
graph = {}

# Loop setiap baris (node asal)
for i in range(len(matrix)):
    graph[nodes[i]] = []  # siapkan list kosong

    # Loop setiap kolom (node tujuan)
    for j in range(len(matrix)):
        # Jika bernilai 1 berarti ada hubungan
        if matrix[i][j] == 1:
            # Tambahkan node tujuan ke list
            graph[nodes[i]].append(nodes[j])

# Menampilkan hasil adjacency list
print("Adjacency List hasil konversi:")
for node, tetangga in graph.items():
    print(node, "->", tetangga)

# Penjelasan:
# Baris A [0,1,1,0] -> A terhubung ke B dan C
# Baris B [1,0,1,0] -> B terhubung ke A dan C
# Baris C [1,1,0,1] -> C terhubung ke A, B, dan D
# Baris D [0,0,1,0] -> D terhubung ke C
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Adjacency List hasil konversi:
A -> ['B', 'C']
B -> ['A', 'C']
C -> ['A', 'B', 'D']
D -> ['C']
PS D:\sem 2\algoritma\Praktikum 11>
```

Pertemua 11_Herbert Batubar_J04403251162

```python
graph = {
    "Andi": ["Budi", "Citra"],
    "Budi": ["Deni"],
    "Citra": ["Andi"],
    "Deni": ["Eka"],
    "Eka": ["Budi"]
}

# Tampilkan node
print("Node:")
for i in graph:
    print(i)

# Tampilkan adjacency list
print("\nAdjacency List:")
for i in graph:
    print(i, "->", graph[i])

# Buat adjacency matrix sederhana
nodes = list(graph.keys())
matrix = []

for i in nodes:
    row = []
    for j in nodes:
        if j in graph[i]:
            row.append(1)
        else:
            row.append(0)
    matrix.append(row)

# Tampilkan adjacency matrix
print("\nAdjacency Matrix:")
for row in matrix:
    print(row)

# Tampilkan hubungan
print("\nHubungan:")
for i in graph:
    for j in graph[i]:
        print(i, "follow", j)
```

```
Node:
Andi
Budi
Citra
Deni
Eka

Adjacency List:
Andi -> ['Budi', 'Citra']
Budi -> ['Deni']
Citra -> ['Andi']
Deni -> ['Eka']
Eka -> ['Budi']

Adjacency Matrix:
[0, 1, 1, 0, 0]
[0, 0, 0, 1, 0]
[1, 0, 0, 0, 0]
[0, 0, 0, 0, 1]
[0, 1, 0, 0, 0]

Hubungan:
Andi follow Budi
Andi follow Citra
Budi follow Deni
Citra follow Andi
Deni follow Eka
Eka follow Budi
PS D:\sem 2\algoritma\Praktikum 11>
```

Andi ---> Budi

| |

v v

Citra Deni ---> Eka

^ |

|_____|

Node (Vertex)

Node merepresentasikan pengguna media sosial, yaitu:

- Andi
- Budi
- Citra
- Deni

- Eka

Edge

Edge merepresentasikan hubungan follow antar pengguna, yaitu:

- Andi → Budi
- Andi → Citra
- Budi → Deni
- Citra → Andi
- Deni → Eka
- Eka → Budi

Jumlah:

- Node = 5
- Edge = 6